

Ex. No. : 02

Date :

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Simple Calculator

Aim

Develop an application to change the font and color of the text and display a toast message when the user presses the button.

Procedure:

Step 1 : File → New Project

Provide the application name (e.g., "Calculator") and click “Next”.

Step 2 : Select the target Android devices

Select the minimum SDK to run the application. Click “Next”.

Step 3 : Choose the activity for the application

By default, choose “Blank Activity”. Click “Next”.

Step 4 : Enter activity name and click “Finish”.

Step 5 : Edit the program

Design a calculator UI in activity_main.xml using Buttons and TextViews.

Implement calculator logic in MainActivity.kt (eg.Addition,Subtractions.,)

Step 6: Run the application

Two ways to run the application:

1. Running through emulator
2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.CALCULATOR"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#3F51B5"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/titleTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#5C6BC0"
        android:gravity="center"
        android:padding="8dp"
        android:text="Simple Calculator"
        android:textColor="#FFFFFF"
        android:textSize="24sp"
        android:textStyle="bold" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2"
        android:background="#FFFFFF"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:id="@+id/expressionTextView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:ellipsize="end"
```

```

        android:gravity="end"
        android:maxLines="2"
        android:padding="8dp"
        android:textColor="#757575"
        android:textSize="22sp" />

<TextView
    android:id="@+id/resultTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ellipsize="end"
    android:gravity="end"
    android:maxLines="1"
    android:padding="8dp"
    android:textColor="#212121"
    android:textSize="36sp"
    android:textStyle="bold" />
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="5"
    android:background="#3F51B5"
    android:orientation="vertical"
    android:padding="4dp">

    <!-- Scientific function buttons -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:orientation="horizontal">

        <Button
            android:id="@+id/btnSin"
            android:layout_width="0dp"

```

```
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#5C6BC0"
android:text="sin"
android:textColor="#FFFFFF"
android:textSize="18sp" />
```

<Button

```
android:id="@+id/btnCos"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#5C6BC0"
android:text="cos"
android:textColor="#FFFFFF"
android:textSize="18sp" />
```

<Button

```
android:id="@+id/btnTan"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#5C6BC0"
android:text="tan"
android:textColor="#FFFFFF"
android:textSize="18sp" />
```

<Button

```
android:id="@+id/btnLog"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#5C6BC0"
```

```

        android:text="log"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />
</LinearLayout>

<!-- More scientific function buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btnSqrt"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#5C6BC0"
        android:text="√"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />

    <Button
        android:id="@+id/btnPow"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#5C6BC0"
        android:text="x^y"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

    <Button
        android:id="@+id/btnMod"
        android:layout_width="0dp"

```

```

        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#5C6BC0"
        android:text="%"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />

<Button
    android:id="@+id/btnClear"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#FF5252"
    android:text="C"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />
</LinearLayout>

<!-- First row of number buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btn7"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#7986CB"
        android:text="7"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />

```

```

<Button
    android:id="@+id/btn8"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="8"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

<Button
    android:id="@+id/btn9"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="9"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

<Button
    android:id="@+id/btnDivide"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#5C6BC0"
    android:text="/"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />
</LinearLayout>

<!-- Second row of number buttons -->
<LinearLayout

```



```

android:layout_width="match_parent"
android:layout_height="0dp"
android:layout_weight="1"
android:orientation="horizontal">

<Button
    android:id="@+id/btn4"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="4"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

<Button
    android:id="@+id/btn5"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="5"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

<Button
    android:id="@+id/btn6"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="6"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

```

```

<Button
    android:id="@+id/btnMultiply"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#5C6BC0"
    android:text="*"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />
</LinearLayout>

<!-- Third row of number buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btn1"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#7986CB"
        android:text="1"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />

    <Button
        android:id="@+id/btn2"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"

```

```

        android:backgroundTint="#7986CB"
        android:text="2"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />

<Button
    android:id="@+id/btn3"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#7986CB"
    android:text="3"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />

<Button
    android:id="@+id/btnSubtract"
    android:layout_width="0dp"
    android:layout_height="match_parent"
    android:layout_margin="2dp"
    android:layout_weight="1"
    android:backgroundTint="#5C6BC0"
    android:text="-"
    android:textColor="#FFFFFF"
    android:textSize="24sp" />
</LinearLayout>

<!-- Fourth row of buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btnDecimal"

```

```
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#7986CB"
android:text="."
android:textColor="#FFFFFF"
android:textSize="24sp" />
```

<Button

```
android:id="@+id/btn0"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#7986CB"
android:text="0"
android:textColor="#FFFFFF"
android:textSize="24sp" />
```

<Button

```
android:id="@+id/btnNegate"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
android:layout_weight="1"
android:backgroundTint="#7986CB"
android:text="+/-"
android:textColor="#FFFFFF"
android:textSize="18sp" />
```

<Button

```
android:id="@+id/btnAdd"
android:layout_width="0dp"
android:layout_height="match_parent"
android:layout_margin="2dp"
```

```

        android:layout_weight="1"
        android:backgroundTint="#5C6BC0"
        android:text="+"
        android:textColor="#FFFFFF"
        android:textSize="24sp" />
</LinearLayout>

<!-- Equals and Delete buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btnDelete"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="1"
        android:backgroundTint="#FFA726"
        android:text="DEL"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

    <Button
        android:id="@+id/btnEquals"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_margin="2dp"
        android:layout_weight="3"
        android:backgroundTint="#4CAF50"
        android:text="="
        android:textColor="#FFFFFF"

```

```
        android:textSize="24sp" />
    </LinearLayout>
</LinearLayout>
</LinearLayout>
```

MainActivity.kt

```

package com.example.calculator

import android.os.Bundle
import android.view.View
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import android.appcompat.app.AppCompatActivity
import kotlin.math.*

class MainActivity : AppCompatActivity() {

    // Use lazy initialization to ensure views are found after setContentView

    private val resultTextView by lazy {
        findViewById<TextView>(R.id.resultTextView) }

    private val expressionTextView by lazy {
        findViewById<TextView>(R.id.expressionTextView) }

    private var lastNumber: Double = 0.0
    private var currentOperation: String = ""
    private var isNewOperation: Boolean = true

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Views are automatically initialized through lazy properties

        // Initialize number buttons
        setupNumberButtons()

        // Setup operation buttons
        setupOperationButtons()
    }

```

```

        // Setup special function buttons
        setupFunctionButtons()

        // Setup other buttons
        setupOtherButtons()
    }

    private fun setupNumberButtons() {
        val numberButtons = arrayOf(
            findViewById<Button>(R.id.btn0),
            findViewById<Button>(R.id.btn1),
            findViewById<Button>(R.id.btn2),
            findViewById<Button>(R.id.btn3),
            findViewById<Button>(R.id.btn4),
            findViewById<Button>(R.id.btn5),
            findViewById<Button>(R.id.btn6),
            findViewById<Button>(R.id.btn7),
            findViewById<Button>(R.id.btn8),
            findViewById<Button>(R.id.btn9)
        )

        numberButtons.forEach { button ->
            button.setOnClickListener {
                if (isNewOperation) {
                    resultTextView.text = ""
                    isNewOperation = false
                }

                val numberText = resultTextView.text.toString()
                val buttonText = button.text.toString()
            }
        }
    }

```



```

        resultTextView.text = numberText + buttonText
        updateExpressionView()
    }
}

// Setup decimal point button
findViewById<Button>(R.id.btnDecimal).setOnClickListener {
    if (isNewOperation) {
        resultTextView.text = "0."
        isNewOperation = false
    } else if (!resultTextView.text.contains(".")) {
        resultTextView.text = resultTextView.text.toString() + "."
    }
    updateExpressionView()
}
}

private fun setupOperationButtons() {
    // Basic arithmetic operations
    val operationButtons = mapOf(
        R.id.btnAdd to "+",
        R.id.btnSubtract to "-",
        R.id.btnMultiply to "*",
        R.id.btnDivide to "/",
        R.id.btnMod to "%"
    )

    operationButtons.forEach { (id, operator) ->
        findViewById<Button>(id).setOnClickListener {
            if (resultTextView.text.isNotEmpty()) {
                lastNumber = resultTextView.text.toString().toDouble()
            }
        }
    }
}

```

```

        currentOperation = operator
        isNewOperation = true
        updateExpressionView()
    }
}

// Equals button
findViewById<Button>(R.id.btnEquals).setOnClickListener {
    if (resultTextView.text.isNotEmpty() &&
        currentOperation.isNotEmpty()) {
        val secondNumber = resultTextView.text.toString().toDouble()
        val result = performOperation(lastNumber, secondNumber,
            currentOperation)

        resultTextView.text = formatResult(result)
        expressionTextView.text = formatResult(lastNumber) + " " +
            currentOperation + " " + formatResult(secondNumber) + " = " +
            formatResult(result)

        lastNumber = result
        currentOperation = ""
        isNewOperation = true
    }
}

private fun setupFunctionButtons() {
    // Scientific functions
    findViewById<Button>(R.id.btnSin).setOnClickListener {
        applyFunction("sin")
    }
    findViewById<Button>(R.id.btnCos).setOnClickListener {
        applyFunction("cos")
    }
}

```

```

        findViewById<Button>(R.id.btnTan).setOnClickListener {
            applyFunction("tan") }

        findViewById<Button>(R.id.btnLog).setOnClickListener {
            applyFunction("log") }

        findViewById<Button>(R.id.btnSqrt).setOnClickListener {
            applyFunction("sqrt") }

        findViewById<Button>(R.id.btnPow).setOnClickListener {
            if (resultTextView.text.isNotEmpty()) {
                lastNumber = resultTextView.text.toString().toDouble()
                currentOperation = "^"
                isNewOperation = true
                updateExpressionView()
            }
        }
    }

private fun setupOtherButtons() {
    // Clear button
    findViewById<Button>(R.id.btnClear).setOnClickListener {
        resultTextView.text = ""
        expressionTextView.text = ""
        lastNumber = 0.0
        currentOperation = ""
        isNewOperation = true
    }

    // Delete button (backspace)
    findViewById<Button>(R.id.btnDelete).setOnClickListener {
        val text = resultTextView.text.toString()
        if (text.isNotEmpty()) {
            resultTextView.text = text.substring(0, text.length - 1)
            updateExpressionView()
        }
    }
}

```

```

    }

    // +/- button (change sign)
    findViewById<Button>(R.id.btnNegate).setOnClickListener {
        if (resultTextView.text.isNotEmpty()) {
            val value = resultTextView.text.toString().toDouble() * -1
            resultTextView.text = formatResult(value)
            updateExpressionView()
        }
    }
}

private fun applyFunction(function: String) {
    if (resultTextView.text.isNotEmpty()) {
        val value = resultTextView.text.toString().toDouble()
        val result = when (function) {
            "sin" -> sin(Math.toRadians(value))
            "cos" -> cos(Math.toRadians(value))
            "tan" -> tan(Math.toRadians(value))
            "log" -> log10(value)
            "sqrt" -> sqrt(value)
            else -> value
        }

        resultTextView.text = formatResult(result)
        expressionTextView.text = "$function($value) = ${formatResult(result)}"

        lastNumber = result
        isNewOperation = true
    } else {
        Toast.makeText(this, "Enter a number first",
            Toast.LENGTH_SHORT).show()
    }
}

```

```

    }
}

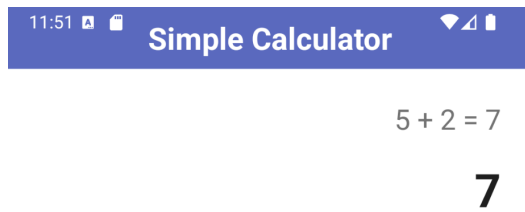
private fun performOperation(first: Double, second: Double, operation:
String): Double {
    return when (operation) {
        "+" -> first + second
        "-" -> first - second
        "*" -> first * second
        "/" -> first / second
        "%" -> first % second
        "^" -> first.pow(second)
        else -> second
    }
}

private fun formatResult(result: Double): String {
    return if (result == result.toInt().toDouble()) {
        result.toInt().toString()
    } else {
        result.toString()
    }
}

private fun updateExpressionView() {
    if (currentOperation.isEmpty()) {
        expressionTextView.text = resultTextView.text
    } else {
        expressionTextView.text = "${formatResult(lastNumber)}
$currentOperation ${resultTextView.text}"
    }
}
}}

```

Output



Result:

The Calculator application successfully takes input, performs arithmetic operations, and displays results correctly on an emulator or mobile device.